FIRST-ORDER LIFTS OF AZUMAYA ALGEBRA ENDOMORPHISMS ARE EQUIVALENT TO PRESERVATION OF THE INDUCED CENTRAL POISSON BRACKET

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Allowed standard prerequisites (editorial).

- Automorphic tensor products, induction/base change and analytic Rankin-Selberg theory. Original complete bibliography and source-local auxiliary L-function arguments retained.

1. INTRODUCTION

A fundamental theme in deformation quantization is the interplay between a noncommutative algebra and the Poisson geometry of its semiclassical limit. In [1], Belov-Kanel and Kontsevich conjectured that the automorphism group of the n-th Weyl algebra $A_n(\mathbb{C})$ is canonically isomorphic to the group of polynomial symplectomorphisms of $\mathbb{C}^{2n}$, equipped with its standard Poisson structure. Their key technique proceeds by reducing the Weyl algebra to positive characteristic p, where $A_n(k)$ becomes an Azumaya algebra whose center $Z \cong k[x_1^p, \dots, x_n^p, \partial_1^p, \dots, \partial_n^p]$ is a polynomial algebra. Here $A_n(k)$ admits a canonical (flat) lift to $A_n(W_2(k))$ — the Weyl algebra over the Witt vectors $W_2(k)$ of length two. The existence of a lift of a Weyl algebra endomorphism f of $A_n(k)$ to $A_n(W_2(k))$ shows that the f is a symplectomorphism when restricted to the center. Belov-Kanel and Kontsevich exploit this structure to construct a homomorphism from the automorphism group of the Weyl algebra to the group of Poisson automorphisms of the center. The same circle of ideas, developed independently by Tsuchimoto [5], also establishes the stable equivalence of the Jacobian and Dixmier conjectures [2].

In [4] we work entirely over a perfect positive characteristic field k and show that a k -algebra endorphism of $A_n(k)$ admits a lift to $A_n(W_2(k))$ if and only if it preserves the canonical Poisson bracket on the center. The goal of this paper is to generalize that result in the framework of *Hochschild cohomology*.

We work in the following general setup. For any field k and any k-algebra A, an endomorphism f of A together with a first-order flat lift of A determines a canonical class in Hochschild cohomology of the bimodule A twisted by f from the left and right; after choosing an R-linear lift of f, this class is represented by a natural 2-cocycle. This cohomology class vanishes if and only if the endomorphism lifts to an endomorphism of the lift of A. Our main observation is that if f preserves the center and the Poisson bracket on the center given by the lift of A, then the restriction to the center of the associated 2-cocycle is symmetric.

If A is an Azumaya algebra of constant rank over a formally smooth center Z in this setup, we prove in Theorem 4.6 that f admits a lift if and only if f preserves the Poisson bracket on Z. The condition $f(Z) \subseteq Z$ is automatic in this case — see Proposition 4.4 for the general statement.

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2. Notation and preliminaries

In the following, k will denote a field and A a k-algebra. We denote by A^{o} the opposite algebra of A and put $A^e = A \otimes_k A^{\mathrm{o}}$. The bar complex of A is the complex $(B_n(A), d_n)$ of free A^e-modules given by $B_n(A) = A^{\otimes_k (n+2)}$ and

$$d_n(a_0 \otimes a_1 \otimes \cdots \otimes a_n \otimes a_{n+1}) = \sum_{i=0}^{n} (-1)^i a_0 \otimes a_1 \otimes \cdots \otimes a_i a_{i+1} \otimes \cdots \otimes a_{n+1}$$

for $n \geq 0$. It is a free resolution of A by the multiplication map $B_0(A) \to A$. For an A-bimodule M, there is a canonical k-isomorphism

$$\mathrm{Hom}_{A^e}(B_n(A), M) \to \mathrm{Hom}_k(A^{\otimes_k n}, M) =: C^n(A/k, M) \tag{2.0.1}$$

given by $h \mapsto (a_1 \otimes \cdots \otimes a_n \mapsto h(1 \otimes a_1 \otimes \cdots \otimes a_n \otimes 1))$ for $n \geq 0$. The complex $C^\bullet(A/k, M)$ is called the *Hochschild complex* of M. It has differentials given by

$$\begin{aligned}(\delta^n f)(a_1 \otimes \cdots \otimes a_{n+1}) = {} & a_1 \cdot f(a_2 \otimes \cdots \otimes a_{n+1}) \\ & + \sum_{i=1}^{n} (-1)^i f(a_1 \otimes \cdots \otimes a_i a_{i+1} \otimes \cdots \otimes a_{n+1}) \\ & + (-1)^{n+1} f(a_1 \otimes \cdots \otimes a_n) \cdot a_{n+1}\end{aligned}$$

for $n \geq 0$. The n-th Hochschild cohomology of A with coefficients in M is defined as

$$\mathrm{HH}^n(A/k, M) := H^n(C^\bullet(A/k, M))$$

for $n \geq 0$. The isomorphisms (2.0.1) is a morphism of complexes and we have the following result.

Proposition 2.1.

$$\mathrm{HH}^i(A/k, M) = \mathrm{Ext}^i_{A^e}(A, M).$$

We call an A-bimodule M, B-diagonal if $bm = mb$ for every $b \in B$, where B is a subalgebra of A.

3. First order flat lifts of endomorphisms

Let R be a commutative local ring with non-zero principal maximal ideal pR such that $p^2 = 0$, and let $k = R/pR$ be its residue field. The map $k \to pR$ given by $r + pR \mapsto pr$ is a well-defined isomorphism; we denote its inverse by $p^{-1} : pR \to k$.

Let A be a k-algebra. By a first-order flat lift of A we mean a flat R-algebra $\widetilde{A}$ together with an identification $\widetilde{A} \otimes_R k \cong A$. We let $\pi : \widetilde{A} \to A$ denote the reduction map, and for $a \in A$ we write $\tilde{a} \in \widetilde{A}$ for an arbitrary lift of a. Since $p^2 = 0$, the $\widetilde{A}$-bimodule $p\widetilde{A} = \ker(\pi)$ is annihilated by p, so its left and right $\widetilde{A}$-actions factor through $A = \widetilde{A}/p\widetilde{A}$; hence $p\widetilde{A}$ is naturally an A-bimodule. Flatness ensures that multiplication by p induces an A-bimodule isomorphism between A and $p\widetilde{A}$. Explicitly, its inverse is given by:

$$p^{-1} : p\widetilde{A} \longrightarrow A, \quad p\tilde{a} \longmapsto \pi(\tilde{a}) = a,$$

and we denote by $\iota : A \to p\widetilde{A}$ the inverse map $a \mapsto p\tilde{a}$.

Remark 3.1. Since R is artinian local, every flat R-module is free. In particular, $\widetilde{A}$ is a free R-module. In fact, if $\{e_i\}_{i \in I}$ is a k-basis for A, then $\{\tilde{e}_i\}_{i \in I}$ forms an R-basis for $\widetilde{A}$.

Now, consider a k-algebra homomorphism $f : A \to A$ and an R-linear first order lift $\widetilde{f} : \widetilde{A} \to \widetilde{A}$ (satisfying $\pi \circ \widetilde{f} = f \circ \pi$). We define the multiplicative defect $D : \widetilde{A} \times \widetilde{A} \to \widetilde{A}$ by:

$$D(x, y) = \widetilde{f}(x)\widetilde{f}(y) - \widetilde{f}(xy).$$

Because $p^2 = 0$ and $\widetilde{f}$ is R-linear, the defect is invariant under adding elements of $p\widetilde{A}$. Consequently, D factors through the reduction map, yielding a well-defined map $L_{\widetilde{f}} : A \otimes_k A \to p\widetilde{A}$ given by:

$$L_{\widetilde{f}}(a \otimes b) := D(\tilde{a}, \tilde{b}).$$

Thus, $\widetilde{f}$ is multiplicative (and hence an R-algebra homomorphism) if and only if $L_{\widetilde{f}}(a \otimes b) = 0$ for all $a, b \in A$. If $\widetilde{f}$ is multiplicative, then automatically $\widetilde{f}(1) = 1$.

Proposition 3.2. *Let $f : A \to A$ be a k-algebra homomorphism and $\widetilde{f} : \widetilde{A} \to \widetilde{A}$ an R-linear lift of f. The k-linear map $C_{\widetilde{f}} \in \mathrm{Hom}_k(A \otimes_k A, A)$ defined by*

$$C_{\widetilde{f}}(a \otimes b) := p^{-1} L_{\widetilde{f}}(a \otimes b)$$

is a 2-cocycle in $C^2(A/k, M)$, where $M = A$ viewed as an A-bimodule via the twisted action

$$a \cdot m \cdot b := f(a) m f(b)$$

for $m \in M$ and $a, b \in A$.

The cohomology class $[C_{\widetilde{f}}] \in H^2(A/k, M)$ is independent of the choice of the R-linear lift $\widetilde{f}$, and $[C_{\widetilde{f}}] = 0$ if and only if f admits a lift to an R-algebra homomorphism $\widehat{f} : \widetilde{A} \to \widetilde{A}$.

Proof. By the definition of the Hochschild differential for the twisted bimodule M, we have for all $a, b, c \in A$:

$$\begin{aligned}(\delta^2 C_{\widetilde{f}})(a \otimes b \otimes c) &= f(a) C_{\widetilde{f}}(b \otimes c) - C_{\widetilde{f}}(ab \otimes c) + C_{\widetilde{f}}(a \otimes bc) - C_{\widetilde{f}}(a \otimes b) f(c) \\ &= p^{-1}\Big(\widetilde{f}(\tilde{a}) L_{\widetilde{f}}(b \otimes c) - L_{\widetilde{f}}(ab \otimes c) + L_{\widetilde{f}}(a \otimes bc) - L_{\widetilde{f}}(a \otimes b) \widetilde{f}(\tilde{c}) \Big).\end{aligned}$$

Expanding $L_{\widetilde{f}}$ via the multiplicative defect D, the term inside the parenthesis becomes:

$$\begin{aligned}&\widetilde{f}(\tilde{a})\big(\widetilde{f}(\tilde{b})\widetilde{f}(\tilde{c}) - \widetilde{f}(\tilde{b}\tilde{c})\big) - \big(\widetilde{f}(\tilde{a}\tilde{b})\widetilde{f}(\tilde{c}) - \widetilde{f}(\tilde{a}\tilde{b}\tilde{c})\big) \\ &+ \big(\widetilde{f}(\tilde{a})\widetilde{f}(\tilde{b}\tilde{c}) - \widetilde{f}(\tilde{a}\tilde{b}\tilde{c})\big) - \big(\widetilde{f}(\tilde{a})\widetilde{f}(\tilde{b}) - \widetilde{f}(\tilde{a}\tilde{b})\big)\widetilde{f}(\tilde{c}).\end{aligned}$$

Due to the associativity of the algebra $\widetilde{A}$, all terms cancel pairwise, yielding $(\delta^2 C_{\widetilde{f}})(a \otimes b \otimes c) = 0$. Thus, $C_{\widetilde{f}}$ is a 2-cocycle.

Now, suppose $\widetilde{f}_1$ and $\widetilde{f}_2$ are two R-linear lifts of f. Their difference $\widetilde{f}_2 - \widetilde{f}_1$ maps into $p\widetilde{A}$ and therefore defines a k-linear map $h : A \to A$ via:

$$(\widetilde{f}_2 - \widetilde{f}_1)(\tilde{a}) = \iota(h(a)).$$

Direct substitution shows that

$$C_{\widetilde{f}_2}(a \otimes b) - C_{\widetilde{f}_1}(a \otimes b) = f(a)h(b) - h(ab) + h(a)f(b) = (\delta^1 h)(a \otimes b).$$

Hence, $C_{\widetilde{f}_2} - C_{\widetilde{f}_1}$ is a coboundary, meaning the cohomology class $[C_{\widetilde{f}}]$ is independent of the lift.

Finally, if f lifts to an R-algebra homomorphism $\widehat{f} : \widetilde{A} \to \widetilde{A}$, then $L_{\widehat{f}} = 0$, making $C_{\widehat{f}} = 0$ and thus $[C_{\widehat{f}}] = 0$. Conversely, suppose $[C_{\widetilde{f}}] = 0$. Then $C_{\widetilde{f}} = \delta^1 h$ for some 1-cochain $h \in \mathrm{Hom}_k(A, A)$, meaning:

$$C_{\widetilde{f}}(a \otimes b) = f(a)h(b) - h(ab) + h(a)f(b) \quad \forall a, b \in A.$$

We define a new R-linear lift $\widehat{f} : \widetilde{A} \to \widetilde{A}$ by correcting $\widetilde{f}$ with h:

$$\widehat{f}(x) := \widetilde{f}(x) - \iota(h(\pi(x))).$$

One readily verifies that $L_{\widehat{f}}(a \otimes b) = 0$ for all $a, b \in A$, proving that $\widehat{f}$ is the desired R-algebra homomorphism. □

It is well known that a first order flat lift of A gives rise to a Poisson bracket on the center of A. This is very helpful in studying the cocycle defined in Proposition 3.2.

3.1. **Restriction of the cocycle to the center.** Let Z be the center of A. For $z_1, z_2 \in Z$,

$$\{z_1, z_2\} = p^{-1}([\tilde{z}_1, \tilde{z}_2]) = p^{-1}(\tilde{z}_1\tilde{z}_2 - \tilde{z}_2\tilde{z}_1)$$

defines a Poisson bracket on the center Z of A.

Proposition 3.3. *Assume that $f(Z) \subseteq Z$ and let $\widetilde{f} : \widetilde{A} \to \widetilde{A}$ be an R-linear lift of f. Let $c \in C^2(Z/k, M)$ be the restriction of $C_{\widetilde{f}}$ to $Z^{\otimes_k 2}$. Then c is a 2-cocycle and*

$$c(x \otimes y) - c(y \otimes x) = \{f(x), f(y)\} - f(\{x, y\})$$

for $x, y \in Z$.

In particular, the induced endomorphism $f : Z \to Z$ preserves the Poisson bracket if and only if the 2-cocycle $c \in C^2(Z, M)$ is symmetric, i.e., $c(x \otimes y) = c(y \otimes x)$ for every $x, y \in Z$.

Proof. Since the Hochschild differential commutes with restriction along $Z \hookrightarrow A$, the restriction c of $C_{\widetilde{f}}$ is again a 2-cocycle.

For $x, y \in Z$,

$$\begin{aligned}
&c(x \otimes y) - c(y \otimes x) = \\
&p^{-1}(\widetilde{f}(\tilde{x})\widetilde{f}(\tilde{y}) - \widetilde{f}(\tilde{x}\tilde{y}) - \widetilde{f}(\tilde{y})\widetilde{f}(\tilde{x}) + \widetilde{f}(\tilde{y}\tilde{x})) = \\
&p^{-1}(\widetilde{f}(\tilde{x})\widetilde{f}(\tilde{y}) - \widetilde{f}(\tilde{y})\widetilde{f}(\tilde{x}) - \widetilde{f}(\tilde{x}\tilde{y} - \tilde{y}\tilde{x})) = \\
&\{f(x), f(y)\} - f(\{x, y\})
\end{aligned}$$

This shows in particular that the antisymmetrization $c(x \otimes y) - c(y \otimes x)$ depends only on f and the lift $\widetilde{A}$, and not on the chosen R-linear lift $\widetilde{f}$. □

Under favorable conditions, symmetric cocycles have trivial cohomology class in the Hochschild cohomology.

Lemma 3.4. *Let Z/k be a commutative algebra which is formally smooth over k in the category of commutative k-algebras, and let M be a diagonal Z-bimodule. Then every symmetric 2-cocycle in the Hochschild complex $C^\bullet(Z/k, M)$ is a coboundary.*

Proof. We interpret $\mathrm{HH}^2(Z/k, M)$ in terms of square-zero extensions. Let $\phi \in \mathrm{Hom}_k(Z^{\otimes 2}, M)$ be a 2-cocycle. This defines a k-algebra structure on the vector space direct sum $E = Z \oplus M$, with multiplication given by

$$(a, m) \cdot (b, n) = (ab, an + mb + \phi(a \otimes b)).$$

Because ϕ is a Hochschild 2-cocycle, E is an associative algebra, and the projection $E \to Z$ is a surjective k-algebra homomorphism whose kernel M satisfies $M^2 = 0$.

Since ϕ is symmetric (i.e., $\phi(a \otimes b) = \phi(b \otimes a)$) and M is a diagonal Z-bimodule (so $am = ma$ for all $a \in Z$ and $m \in M$), the multiplication on E is commutative. Thus, E is a commutative square-zero extension of Z by M.

By definition, the k-algebra Z is formally smooth if and only if every extension of Z by a nilpotent ideal in the category of commutative k-algebras splits. Therefore, there exists a k-algebra section $s \colon Z \to E$. This section must be of the form $s(a) = (a, \eta(a))$ for some k-linear map $\eta : Z \to M$. Preserving multiplication requires

$$(ab, \eta(ab)) = s(ab) = s(a)s(b) = (a, \eta(a))(b, \eta(b)) = (ab, a\eta(b) + \eta(a)b + \phi(a \otimes b)),$$

which rearranges to

$$\phi(a\otimes b)=\eta(ab)-a\eta(b)-\eta(a)b.$$

This is precisely the statement that $\phi=-d\eta$ in the Hochschild complex, meaning ϕ is a coboundary. □

4. Lifts of endomorphisms of Azumaya algebras

If A is a k-algebra and at the same time an Azumaya algebra over its center Z, then the multiplication map $\mu : A\otimes_Z A^{\mathrm{o}}\to A$ admits an A^e-module splitting s and A is a finitely generated projective Z-module (see [3, III, Théorème 5.1, Théorème 1.4]).

Lemma 4.1. *If A is an Azumaya algebra with center Z and M is an A-bimodule, then the restriction map $C^i(A/k,M)\to C^i(Z/k,M)$ induces an injective map*

$$\mathrm{HH}^i(A/k,M)\hookrightarrow \mathrm{HH}^i(Z/k,M)$$

on Hochschild cohomology.

Proof. Let $\Lambda=A\otimes_Z A^{\mathrm{o}}$. The complex $P_\bullet:=A^e\otimes_{Z^e}B_\bullet(Z/k)$ is a projective resolution of $A^e\otimes_{Z^e}Z\cong\Lambda$. The multiplication map $\mu:\Lambda\to A$ lifts to an A^e-chain map $r_\bullet : P_\bullet\to B_\bullet(A/k)$ given by

$$r_n((a\otimes b^{\mathrm{o}})\otimes z_0\otimes\cdots\otimes z_{n+1})=az_0\otimes\cdots\otimes z_{n+1}b.$$

By adjunction, we have an isomorphism of complexes

$$\mathrm{Hom}_{A^e}(P_\bullet,M)\cong\mathrm{Hom}_{Z^e}(B_\bullet(Z/k),M)=C^\bullet(Z,M),$$

which upon taking cohomology yields a canonical isomorphism $\mathrm{Ext}^i_{A^e}(\Lambda,M)\cong\mathrm{HH}^i(Z/k,M)$.

Under this identification, the map induced by $r_\bullet^*$ on cohomology corresponds precisely to the restriction map $\mathrm{HH}^i(A/k,M)\to\mathrm{HH}^i(Z/k,M)$ and $r_\bullet^*$ computes the pullback $\mu^*:\mathrm{Ext}^i_{A^e}(A,M)\to\mathrm{Ext}^i_{A^e}(\Lambda,M)$ for $i\geq 0$.

Since A is an Azumaya algebra, μ admits an A^e-module splitting s such that $\mu s=\mathrm{id}_A$. Therefore $s^*\mu^*=\mathrm{id}$, making μ^*—and consequently the restriction map—injective. □

Remark 4.2. Suppose M is a Z-diagonal and $s(1)=\sum_i x_i\otimes y_i\in A\otimes_Z A^{\mathrm{o}}$. Then the map $e_M:M\to M$ given by $e_M(m)=\sum_i x_i m y_i$ is a well-defined Z-module projection onto $M^A=\{m\in M\mid am=ma,\text{for every } a\in A\}$. Then one may prove that the restriction map, followed by the homomorphism induced by e_M, yields an isomorphism

$$\mathrm{HH}^i(A/k,M)\xrightarrow{\sim}\mathrm{HH}^i(Z/k,M^A)$$

on Hochschild cohomology. Let σ be a lift of $s(1)=\sum_i x_i\otimes y_i\in A\otimes_Z A^{\mathrm{o}}$ in A^e, and let $e=s\mu$. Then e is right multiplication by σ, and the induced chain map on $P_\bullet$ is

$$e_n(u\otimes\xi)=u\sigma\otimes\xi.$$

Under the adjunction map, precomposition with e_n corresponds to postcomposition with e_M on Hochschild cochains $C^n(Z,M)$. Since e_M is a split idempotent with image M^A, the image of the induced endomorphism on $C^\bullet(Z,M)$ is the direct summand subcomplex $C^\bullet(Z,M^A)$. Passing to cohomology yields the claimed isomorphism.

We now prove that a unital *ring* endomorphism f of an Azumaya algebra of constant rank over an *arbitrary* commutative ring Z satisfies $f(Z)\subseteq Z$. We have only been able to find this result in the literature for reduced rings and therefore give a complete proof below. First we need the following lemma.

Lemma 4.3. *Let B be an Azumaya algebra of degree n over a commutative ring C, let k be a field, and let*

$$u : B \to S$$

be a ring homomorphism into a central simple k-algebra S of degree n. Then the k-subalgebra of S generated by $u(B)$ is all of S.

Proof. Let T be the k-subalgebra of S generated by $u(B)$. Suppose $T \neq S$. Choose a simple quotient

$$T \twoheadrightarrow S_0,$$

and put $K = Z(S_0)$. The image of C in S_0 is central, hence lies in K. Therefore the composite $B \to S_0$ induces a K-algebra map

$$\theta : B \otimes_C K \to S_0.$$

Its image is the K-subalgebra generated by the image of B, hence all of S_0.

Now $B \otimes_C K$ is central simple of degree n over K, so a nonzero surjective homomorphism from it is an isomorphism. Thus

$$\dim_k S_0 = \dim_k (B \otimes_C K) = n^2 [K : k] \geq n^2$$

contradicting that $\dim_k S_0 < \dim_k S = n^2$. Hence $T = S$. □

Proposition 4.4. *Let A be an Azumaya algebra of constant degree over a commutative ring R, and let*

$$f : A \to A$$

be a ring endomorphism. Then $f(R) \subseteq R$.

Proof. Let

$$N := \sum_{a \in A} R f(a) \subseteq A$$

be the R-submodule generated by $f(A)$. It suffices to prove that

$$N = A.$$

Fix a maximal ideal $\mathfrak{m} \subset R$, and set $k(\mathfrak{m}) = R/\mathfrak{m}$ and $S = A/\mathfrak{m}A$. Since A has constant degree n, S is a central simple $k(\mathfrak{m})$-algebra of degree n. Consider the composite

$$\psi_{\mathfrak{m}} : A \xrightarrow{f} A \to S.$$

Let $H = \psi_{\mathfrak{m}}(A) \subseteq S$. The image of N in S is exactly the $k(\mathfrak{m})$-span $k(\mathfrak{m})H$. Since H is a subring, $k(\mathfrak{m})H$ is precisely the $k(\mathfrak{m})$-subalgebra generated by H. By Lemma 4.3,

$$k(\mathfrak{m})H = S.$$

Hence the image of N in $A/\mathfrak{m}A$ is all of $A/\mathfrak{m}A$, i.e. $N + \mathfrak{m}A = A$. Since $\mathfrak{m}$ was arbitrary, we have $(A/N) \otimes_R k(\mathfrak{m}) = 0$ for every maximal ideal $\mathfrak{m} \subset R$. As A/N is a finitely generated R-module, it follows that $A/N = 0$, and hence $N = A$. □

Remark 4.5. The constant-degree hypothesis is necessary. Indeed, let $A = \mathbb{C} \times \mathrm{Mat}_2(\mathbb{C})$. Then the ring endomorphism given by

$$(a, M) \mapsto \left(a, \begin{pmatrix} a & 0 \\ 0 & \overline{a} \end{pmatrix} \right)$$

maps the central element $(i, 0)$ to a non-central element.

Finally we state and prove the main application and result of this paper.

Theorem 4.6. *Let k be a field, A a k-algebra, $f : A \to A$ a k-algebra endomorphism of A and $\widetilde{A}$ a first order flat lift of A. Assume that A is an Azumaya algebra of constant rank over its center Z, and that Z is formally smooth over k in the category of commutative k-algebras. Then f admits a lift to an endomorphism $\widetilde{f} : \widetilde{A} \to \widetilde{A}$ if and only if f preserves the Poisson bracket on Z induced by $\widetilde{A}$.*

Proof. By Proposition 3.2, the endomorphism f together with any R-linear lift $\widetilde{f}$ of f determines a Hochschild 2-cocycle $C_{\widetilde{f}} \in C^2(A/k,M)$, whose cohomology class $[C_{\widetilde{f}}] \in \mathrm{HH}^2(A/k,M)$ vanishes if and only if f admits a multiplicative lift. Recall here that $M = A$ with the bimodule structure $amb = f(a)mf(b)$ for $a,b,m \in A$.

Suppose first that f admits a multiplicative lift $\widehat{f}$. Then for $x,y \in Z$ and lifts $\tilde{x},\tilde{y} \in \widetilde{A}$ we have

$$f(\{x,y\}) = p^{-1}\big(\widehat{f}([\tilde{x},\tilde{y}])\big) = p^{-1}\big([\widehat{f}(\tilde{x}),\widehat{f}(\tilde{y})]\big) = \{f(x),f(y)\},$$

so f preserves the Poisson bracket on Z.

Suppose conversely that f preserves the Poisson bracket on Z. Since A is Azumaya of constant rank, Proposition 4.4 gives $f(Z) \subseteq Z$. Hence M, restricted to Z, is a diagonal Z-bimodule. Proposition 3.3 then shows that the restriction of $C_{\widetilde{f}}$ to Z is a symmetric 2-cocycle in $C^2(Z/k,M)$. Because Z is formally smooth, Lemma 3.4 implies that this restricted cocycle is a coboundary. Finally, the restriction map $\mathrm{HH}^2(A/k,M) \hookrightarrow \mathrm{HH}^2(Z/k,M)$ is injective by Lemma 4.1, so $[C_{\widetilde{f}}] = 0$ in $\mathrm{HH}^2(A/k,M)$, and therefore f lifts. □

Remark 4.7. Let k be a perfect field of positive characteristic and let $A = A_n(k)$ be the n-th Weyl algebra over k. Then it follows from Theorem 4.6, that an endomorphism of A lifts to $A_n(W_2(k))$ if and only if f preserves the canonical Poisson bracket on the center of A. This is exactly [4, Theorem 2.7].

It would be interesting to formulate criteria for the existence of lifts from $W_m(k)$ to $W_{m+1}(k)$ for $m \geq 2$. However, the framework developed in this paper does not seem to extend directly to that setting. Already for $m = 2$, $A_1(W_2(k))$ is no longer an Azumaya algebra, so the present methods do not apply in any immediate way. Nevertheless, we expect that the techniques of [4] can be adapted to treat this problem for the Weyl algebra.

For $p = 2$, the endomorphism

$$x \mapsto x, \qquad \partial \mapsto \partial + x^3\partial^4$$

of $A_1(k)$ lifts to $W_2(k)$, but not to $W_3(k)$. We expect that the general lifting problem from $k = W_1(k)$ to $W_m(k)$ is governed by subtler obstructions, and it seems plausible that the de Rham–Witt complex plays a role in describing them.

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